Problem Set 5

Due Saturday, October 10, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

Reminder: We will be having our midterm 1 exam on Thursday, October 15, 7 - 9 pm in Jefferson 250. It will cover topics 1 and 2 of the syllabus. You are allowed to bring two pages of notes (two sides of one sheet or one side of two sheets, of any font size, typed or hand-written). You will not need a calculator. We will be posting practice problems by Saturday, October 10, 11:59 pm and doing reviews during sections in the exam week. Problem Set 6 will be posted on Saturday, October 17. Problem 1 below is from Topic 2 while problems 2 and 3 are from Topic 3.

1. Consider a particle represented by the Gaussian wave packet $\psi(x,0) = Ae^{-\frac{x^2}{2a^2}}e^{ik_0x}$ in position space at time $t = 0$, where a and k_0 are constants.

   (a) Normalize $\psi(x,0)$. (5 points)

   (b) Determine the wavefunction $\phi(k,0)$ at time $t = 0$ in momentum space using the Fourier transform, $\phi(k,0) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx\,\psi(x,0)e^{-ikx}$. (5 points)

   (c) Check if $\phi(k,0)$ is normalized through the normalization of $\psi(x,0)$ or if we need to normalize again. (5 points)

   (d) Determine $\psi(x,t)$ and show that $|\psi(x,t)|^2$ can be written as $|\psi(x,t)|^2 = \frac{1}{\sqrt{\pi}a(t)}e^{-\frac{(x-\frac{\hbar k_0}{m}t)^2}{a(t)^2}}$, and determine $a(t)$. (5 points)

   (e) Discuss how $|\psi(x,t)|^2$ evolves in time. How long does it take for the width to double in size? How far has the particle traveled during this time? Which Gaussian packet's width doubles in size faster, an initially wider or narrower packet? (5 points)

   (f) Determine the uncertainty in the position of the particle Δx at time t. (5 points)

   (g) Determine the uncertainty in the momentum of the particle Δp at time t. (5 points)

   (h) Write down $\Delta x \Delta p$ at time t and check if it obeys the uncertainty principle. What is the value of $\Delta x \Delta p$ at time $t = 0$ and what happens as t increases and $t \to \infty$? (5 points)

2. (a) Given two state vectors $|\phi\rangle$ and $|\psi\rangle$, prove the Schwartz inequality $|\langle\phi|\psi\rangle|^2 \leq \langle\phi|\phi\rangle\langle\psi|\psi\rangle$. (5 points)

   (b) Given normalized kets $|\alpha\rangle$, $|\beta\rangle$, $|\gamma\rangle$, $|\delta\rangle$ and operators $\hat{A} = |\delta\rangle\langle\gamma|$ and $\hat{B}^\dagger = |\beta\rangle\langle\alpha|$, write and simplify, as much as possible, the matrix element $\left(\langle\alpha|\hat{B}^2\hat{A}^\dagger|\delta\rangle\right)^*$. (5 points)

(c) Show that given a Hermitian operator $\hat{A}$ that $\hat{U} = e^{i\hat{A}}$ is unitary; i.e., $\hat{U}^\dagger \hat{U} = 1$. (5 points)

(d) Given the initial state $|\psi(0)\rangle$ at time $t = 0$ and a time-independent Hamiltonian operator $\hat{H}$, show that $|\psi(t)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}|\psi(0)\rangle$ solves the Schrödinger equation at time t. Does $|\psi(t)\rangle$ stay normalized if the initial state $|\psi(0)\rangle$ is normalized (or is the time evolution operator $e^{-\frac{i}{\hbar}\hat{H}t}$ unitary)? (5 points)

(e) Given two operators $\hat{A}$ and $\hat{B}$, show that $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$. (5 points)

(f) Given two operators $\hat{A}$ and $\hat{B}$, prove that $e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2}[\hat{A}, [\hat{A}, \hat{B}]] + \cdots$ (5 points)

(g) Determine the Hermitian conjugates of the operators $\hat{x}$, i, and $\frac{\partial}{\partial x}$. (5 points)

(h) The Hamiltonian operator of a 2-level system is given by $\hat{H} = \alpha\,(|1\rangle\langle 1| - |2\rangle\langle 2| + i|1\rangle\langle 2| - i|2\rangle\langle 1|)$, where α is a real constant. Show that $\hat{H}$ is Hermitian. Find its eigenvalues and eigenvectors. (5 points)

3. In problem 1 above, we investigated the Gaussian wave packet $\psi(x, t)$ in position representation (or position space), given $\psi(x, 0) = Ae^{-\frac{x^2}{2a^2}} e^{ik_0 x}$ with $A = \frac{1}{\sqrt{\sqrt{\pi}\, a}}$ when normalized. Now do the calculations below in the momentum representation (or momentum space) starting with $\phi(k, 0) = \sqrt{\frac{a}{\sqrt{\pi}}} e^{-\frac{1}{2}a^2(k-k_0)^2}$ that you obtained in problem 1b.

   (a) Determine $\langle\hat{p}\rangle$ and $\langle\hat{p}^2\rangle$ at time t. (5 points)

   (b) Determine $\langle\hat{x}\rangle$ and $\langle\hat{x}^2\rangle$ at time t. (5 points)

   (c) Check that the uncertainty principle is satisfied with the minimum uncertainty saturated at $t = 0$. (5 points)

4. How long did it take you to complete this problem set?